Fiscal Year:	FY 2012	**Task Last Updated:**	FY 09/24/2012
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Flight Support and Data Analysis for the CVB Microgravity Flight Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	07/15/2009	**End Date:**	09/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS Increments 23/24 , 25/26 NOTE: Extended to 9/30/13 per PI and NSSC information (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AL98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Constrained Vapor Bubble Experiment (CVB) is studying the fundamental physics and interfacial phenomena underlying the operation and performance of a miniature, wickless heat pipe in microgravity. The goal of the research is to understand the fluid mechanics and heat transfer processes occurring in these devices as we evaporate and condense the working fluid, pentane. The practical objective of the work is to improve the design of future space-based thermal control systems. In addition, the research will help improve the design of terrestrial heat pipes for use in laptops, servers, high-power LED luminaires, and even for the treatment of epileptic seizures. The CVB is composed of a glass heat pipe fabricated from a conventional spectrophotometer cuvette. An optical technique is used to measure the shape of the vapor-liquid interface over the entire surface of the cuvette. Thermocouples are used to measure the temperature gradient along the main axis of the heat pipe as well as several off-axis points. By measuring the shape of the vapor-liquid interface we calculate the capillary forces driving fluid flow from the cooled to the heated end of the system and by measuring the temperature profile we also calculate the heat transfer throughout the system. Thus we obtain a complete description of the operation of the device and can literally "see" why it operates as it does on Earth and in microgravity. NOTE: this is a continuation of NNC05GA27G and NAG3-2383 (PI=Wayner).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this last experiment will help to improve the design of all devices that rely on changing the phase of a fluid such as boilers, distillation columns, and refrigeration systems.
	Work continues on this experiment on ISS. Several presentations have been made and papers published. See below and Task Book Bibliography for papers published: Presentations [PR]: PR01) Change-of-Phase Heat Transfer on Nanostructured Surfaces, Joel L.Plawsky, Manas Ojha, Arya Chatterjee, and Peter C. Wayner, Jr. 20th International Symposium on Transport Phenomena, 7-10 July, 2009, Victoria, BC Canada PR02) Modeling Contact Line Dynamics in Evaporating Menisci, Joel L.Plawsky, Arya Chatterjee and Peter C. Wayner, Jr. 2009 Comsol User's Conference 8-10 October, Boston , MA USA PR03) The Constrained Vapor Bubble Experiment for ISS – Earth's Gravity Results, Joel L.Plawsky, Arya Chatterjee and Peter C. Wayner, Jr. 2009 AIChE Annual Meeting 8-13 November, Nashville, TN USA PR04) The Constrained Vapor Bubble Experiment for ISS – Earth's Gravity Results, A. Chatterjee, J. L. Plawsky and P. C. Wayner, Jr., American Institute of Aeronautics and Astronautics Conference 2010, Orlando, FL, USA PR05) A Boundary Value Model for the Evaporation Meniscus, A. Chatterjee, J. L. Plawsky and P. C. Wayner, Jr., International Heat Transfer Conference August 2010, Washington DC, USA (IHTC14-22677) PR 06) American Astronautical Society National Conference, Nov. 16-17, 2010, Cape Canaveral, Fl, "Constrained Vapor Bubble", Peter C. Wayner, Jr., Presentations from the AAS National Conference (ISS – The Next Decade) have been posted on the AAS web site www.astronautical.org < http://www.astronautical.org /> PR 07) "The Constrained Vapor Bubble Experiment – Results from the International Space Station", Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. , Ronald J Sicker, David F Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, Presented at the 2010 Annual AIChE Meeting, November 7-12, 2010 Salt Palace Convention Center Salt Lake City, UT. PR 08) "The Constrained Vapor Bubble (CVB) Experiment in the Microgravity Environment of the International Space Station", Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. , Ronald J Sicker, David F Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, 9-page Extended Abstract submitted for the 49th AIAA Aerospace Sciences Meeting, Orlando, Fl, Jan, 2011. Final proceedings paper #AIAA-2011-1197 was presented and published in the proceedings. PR 09) "The Constrained Vapor Bubble Experiment – Results from the International Space Station", Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. , Ronald J Sicker, David F Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, Paper # AJTEC2011 -44015: Draft Submitted for Proceedings of the ASME/JSME 8th Thermal Engineering Joint Conference March 13-17, 2011, Honolulu, Hawaii, USA. PR 10) "The Constrained Vapor Bubble Heat Pipe-On Earth and in Space", Joel L. Plawsky TU Darmstadt, Darmstadt, Germany, October 22, 2010 PR 11) Invited Keynote Talk: "The Constrained Vapor Bubble, CVB, Science and Application on Earth and on the ISS", Peter C. Wayner, Jr., Eurotherm Seminar 92 on Gravitational Effects on Liquid-Vapor Phase Change, Presqu'ile de Giens, France, April 17-21, 2011. PR 12) "The Constrained Vapor Bubble Heat Pipe-On Earth and in Space", Joel L. Plawsky CUNY, New York, NY March 1, 2011. PR 13) Presentation Concerning CVB at International Space Station (ISS) Research Forum, Great Lakes Science Center & NASA Glenn Research Center, Cleveland, OH, June 17, 2011. PR 14) Presentation at International Space Station (ISS) Research Forum, Great Lakes Science Center & NASA Glenn Research Center, Cleveland, OH, June 17, 2011. PR 15) "The CVB Experiment :Constrained Bubble Nucleation in Microgravity", A. Chatterjee, P.C.Wayner, Jr., J.L. Plawsky, David F. Chao , Ronald J. Sicker, Tibor Lorik, L. Chestney, J. Eustace, R. Margie, J. Zoldak, Sixth International Conference on Two Phase Systems for Ground and Space Applications, Cava de' Tirreni, Italy, September 25-28, 2011.

Task Progress:

PR 16) Joel L. Plawsky, Peter C. Wayner, Jr., and Arya Chatterjee; David F. Chao and Ronald J. Sicker; Tibor Lorik , Louis Chestney, Raymond Margee, John Eustace and John Zoldak, " The CVB Heat Pipe Experiment – Explosive Nucleation in the 20 mm Module," AIChE Annual Meeting, Minneapolis, MN November 2011.

PR 17) “The Constrained Vapor Bubble Heat Pipe-On Earth and in Space”, Joel L. Plawsky NJIT, November 22, 2011.

PR 18) American Astronautical Society National Conference, June 25-28, 2012, Denver, CO, “A Transparent Heat Pipe on the ISS: Lessons from the Constrained Vapor Bubble, CVB, Experiment”, J.L. Plawsky, Presentations from the AAS National Conference have been posted on the AAS web site www.astronautical.org < http://www.astronautical.org/ >

PR 19) 10th International Conference on Nanochannels, Microchannels, and Minichannels, ICNMM2012, July 8-12, 2012, Puerto Rico, USA “Heat Pipe Performance In Microgravity: Lessons From The Constrained Vapor Bubble, CVB, Capillary Fin Experiment,” Joel L. Plawsky and Peter C. Wayner, Jr..

Papers Under Review (SU)

SU 1) “The Constrained Vapor Bubble (CVB) Experiment in the Microgravity Environment of the International Space Station,” Arya Chatterjee, Joel L. Plawsky, Peter C. Wayner, Jr., Ronald J. Sicker, David F. Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak and Raymond Margie. Rensselaer Polytechnic Institute, NASA Glenn Research Center, and Zin Technologies. Submitted to the AIAA Journal.

Papers Published [PUB]:

PUB 1) Arya Chatterjee, Joel L. Plawsky, Peter C. Wayner Jr, David F. Chao, Ronald J. Sicker, Tibor Lorik, Louis Chestney, John Eustace, and John Zoldak, “Constrained Vapor Bubble Experiment for International Space Station: Earth’s Gravity Results,” J. Thermophysics and Heat Transfer, 24, 400-408 (2010).

PUB 2) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr., “A Boundary Value Model for the Evaporation Meniscus” 14th International Heat Transfer Conference, Washington, DC, August 8-13, 2010. The paper (# IHTC14-22677) was published in the proceedings.

PUB 3) Manas Ojha, Arya Chatterjee, George Dalakos, Peter C. Wayner, and Joel L. Plawsky, “Role of solid surface structure on evaporative phase change from a completely wetting corner meniscus,” Physics of Fluids, 22, 052101 (2010)

PUB 4) Manas Ojha, Arya Chatterjee, Frank Mont, E.F. Schubert, Peter C. Wayner Jr. and Joel L. Plawsky, “The role of solid surface structure on dropwise phase change processes,” Int. J. Heat and Mass Transfer, 53, 910-922, (2010).

PUB 5) Joel L.Plawsky, Manas Ojha, Arya Chatterjee, and Peter C. Wayner, Jr., "Change-of-Phase Heat Transfer on Nanostructured Surfaces," Proceedings of the 20th International Symposium on Transport Phenomena, 7-10 July, 2009, Victoria, BC Canada

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PUB 9) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. , Ronald J Sicker, David F Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, “The Constrained Vapor Bubble (CVB) Experiment in the Microgravity Environment of the International Space Station”, Proceedings of the 49th AIAA Aerospace Sciences Meeting, Orlando, Fl, Jan, 2011. The paper, #AIAA-2011-1197.

PUB 10) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. , Ronald J Sicker, David F Chao, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, “The Constrained Vapor Bubble Experiment – Results from the International Space Station”, Proceedings of the ASME/JSME 8th Thermal Engineering Joint Conference, March 13-17, 2011, Honolulu, Hawaii, USA. The paper, # AJTEC2011 -44015.

PUB 11) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr., David Chao, Ronald J. Sicker, Tibor Lorik, Louis Chestney, John Eustace, John Zoldak, “The Constrained Vapor Bubble Experiment for ISS- Earth’s Gravity Micro-gravity Results,” Proceedings of the 24th Symposium on Gravity-Related Phenomena in Space Application, 48th AIAA Aerospace Science Meeting, 4-7 January, 2010, Orlando, Fl. The paper, AIAA 2010-1481.

PUB 12) “Station Research Feature, Keeping Cool With Heat Pipes” at: http://www.nasa.gov/

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Thesis (TH)

TH1 Arya Chatterjee, "The Constrained Vapor Bubble on Earth and Space" (December 2010).

LIS Reports

LIS 1) Joel L. Plawsky and Peter C. Wayner, Jr, “30-Day Report for Constrained Vapor Bubble (CVB) Increments 23/24, October 2010.

LIS 2) Joel L. Plawsky and Peter C. Wayner, Jr, “30-Day Report for Constrained Vapor Bubble (CVB) Increments 25/26, April 2011.

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